Jurnal Modul 07
Basis Data
08_Scripting_SQL_Create

Disusun Oleh
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Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawab :

1. desain database
   a. query

```
CREATE TABLE department (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);
```

   output

```
Table DEPARTMENT created.
```

   b. query

```
CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);
```

   output

```
Table EMPLOYEE created.
```

c. tabel berelasi

- table supplier

```sql
CREATE TABLE supplier (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    phone VARCHAR2(20)
);
```

Script Output ×

Task completed in 0.036 seconds

```
Table SUPPLIER created.
```

- table barang

```sql
CREATE TABLE barang (
    barang_id NUMBER PRIMARY KEY,
    barang_name VARCHAR2(100),
    stock NUMBER,
    harga NUMBER,
    supplier_id NUMBER,
    CONSTRAINT fk_supplier
        FOREIGN KEY (supplier_id)
        REFERENCES supplier(supplier_id)
);
```

Script Output ×

Task completed in 0.045 seconds

```
Table BARANG created.
```

2. analisis table

a. query

```sql
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE barang COMPUTE STATISTICS;
```

b. output

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.


Table BARANG analyzed.
```

3. data setiap table
    a. departement

```sql
INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'Finance', 102, 1002);
INSERT INTO department VALUES (30, 'IT', 103, 1003);
INSERT INTO department VALUES (40, 'Marketing', 104, 1004);
INSERT INTO department VALUES (50, 'Production', 105, 1005);
INSERT INTO department VALUES (60, 'Sales', 106, 1006);
INSERT INTO department VALUES (70, 'Warehouse', 107, 1007);
INSERT INTO department VALUES (80, 'Support', 108, 1008);
INSERT INTO department VALUES (90, 'Purchasing', 109, 1009);
INSERT INTO department VALUES (100, 'Security', 110, 1010);
```

    b. employee

```sql
INSERT INTO employee VALUES (1, 'Andi', 'andi@gmail.com', 5000000, 0.1, TO_DATE('2022-01-10','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (2, 'Budi', 'budi@gmail.com', 4500000, 0.1, TO_DATE('2022-02-11','YYYY-MM-DD'), 20);
INSERT INTO employee VALUES (3, 'Citra', 'citra@gmail.com', 7000000, 0.2, TO_DATE('2022-03-12','YYYY-MM-DD'), 30);
INSERT INTO employee VALUES (4, 'Deni', 'deni@gmail.com', 5500000, 0.1, TO_DATE('2022-04-13','YYYY-MM-DD'), 40);
INSERT INTO employee VALUES (5, 'Eka', 'eka@gmail.com', 6000000, 0.2, TO_DATE('2022-05-14','YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (6, 'Fajar', 'fajar@gmail.com', 5200000, 0.1, TO_DATE('2022-06-15','YYYY-MM-DD'), 60);
INSERT INTO employee VALUES (7, 'Gina', 'gina@gmail.com', 5300000, 0.2, TO_DATE('2022-07-16','YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (8, 'Hani', 'hani@gmail.com', 7500000, 0.3, TO_DATE('2022-08-17','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (9, 'Indra', 'indra@gmail.com', 4700000, 0.1, TO_DATE('2022-09-18','YYYY-MM-DD'), 90);
INSERT INTO employee VALUES (10, 'Joko', 'joko@gmail.com', 6500000, 0.2, TO_DATE('2022-10-19','YYYY-MM-DD'), 100);
```

    c. supplier

```sql
INSERT INTO supplier VALUES (1, 'PT Sumber Makmur', '0811111111');
INSERT INTO supplier VALUES (2, 'PT Jaya Abadi', '0822222222');
INSERT INTO supplier VALUES (3, 'PT Sentosa', '0833333333');
INSERT INTO supplier VALUES (4, 'PT Maju Bersama', '0844444444');
INSERT INTO supplier VALUES (5, 'PT Sejahtera', '0855555555');
INSERT INTO supplier VALUES (6, 'PT Amanah', '0866666666');
INSERT INTO supplier VALUES (7, 'PT Teknologi', '0877777777');
INSERT INTO supplier VALUES (8, 'PT Digital', '0888888888');
INSERT INTO supplier VALUES (9, 'PT Nusantara', '0899999999');
INSERT INTO supplier VALUES (10, 'PT Global', '0800000000');
```

    d. barang

```sql
INSERT INTO barang VALUES (101, 'Laptop', 10, 8000000, 1);
INSERT INTO barang VALUES (102, 'Mouse', 50, 150000, 2);
INSERT INTO barang VALUES (103, 'Keyboard', 40, 300000, 3);
INSERT INTO barang VALUES (104, 'Monitor', 20, 2500000, 4);
INSERT INTO barang VALUES (105, 'Printer', 15, 2000000, 5);
INSERT INTO barang VALUES (106, 'Scanner', 12, 1800000, 6);
INSERT INTO barang VALUES (107, 'Speaker', 30, 500000, 7);
INSERT INTO barang VALUES (108, 'Flashdisk', 100, 100000, 8);
INSERT INTO barang VALUES (109, 'Router', 25, 750000, 9);
INSERT INTO barang VALUES (110, 'Webcam', 18, 600000, 10);
```

4. analisis table

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE barang COMPUTE STATISTICS;
```

Script Output ×

Task completed in 0.035 seconds

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.


Table BARANG analyzed.
```

5. view empvu80

```
CREATE VIEW empvu80 AS
SELECT employee_id,
       last_name AS name,
       salary,
       department_id
FROM employee
WHERE department_id = 80;
```

Script Output ×

Task completed in 0.051 seconds

```
View EMPVU80 created.
```

6. struktur view empvu80
   - query

```
SELECT * FROM user_views WHERE view_name = 'EMPVU80';
```

   - output

	VIEW_NAME	TEXT_LENGTH	TEXT
1	EMPVU80	120	SELECT employee_id, last_name AS name, salary, department_idFROM employeeWHERE department_id = 80

	TEXT_VC	TYPE_TEXT_LENGTH	TYPE_TEXT	OID_TEXT_LENGTH	OID_TEXT
1	SELECT employee_id, last_name AS name, salary, department_idFROM employeeWHERE department_id = 80	(null)	(null)	(null)	(null)

	VIEW_TYPE_OWNER	VIEW_TYPE	SUPERVIEW_NAME	EDITIONING_VIEW	READ_ONLY	CONTAINER_DATA	BEQUEATH	ORIGIN_CON_ID	DEFAULT_COLLATION	CONTAINERS_DEFAULT	CONTAINER_MAP
1	(null)	(null)	(null)	N	N	N	DEFINER	1	USING_NLS_COMP	NO	NO

EXTENDED_DATA_LINK	EXTENDED_DATA_LINK_MAP	HAS_SENSITIVE_COLUMN	ADMIT_NULL	PDB_LOCAL_ONLY
NO	NO	NO	NO	NO

7. non-unique index

- query

```sql
CREATE INDEX idx_department_id
ON employee(department_id);

SELECT index_name, table_name
FROM user_indexes
WHERE table_name = 'EMPLOYEE';
```

- output

	INDEX_NAME	TABLE_NAME
1	SYS_C008369	EMPLOYEE
2	IDX_DEPARTMENT_ID	EMPLOYEE